

Report

1. Theoretical Bitrate Calculation

The theoretical bitrate is determined by the LoRa modulation parameters configured in the software (Bw125Cr45Sf128).

Parameters:

- Bandwidth (BW): 125kHz = 125,000Hz
- Spreading Factor (SF): 7 (Derived from $2^7 = 128$ chips/symbol)
- Coding Rate (CR): $4/5 = 0.8$

Formula:

$$\text{Bitrate (R)} = \text{SF} * (\text{BW} / 2^{\text{SF}}) * \text{CR}$$

Calculation:

$$R = 7 * (125,000 / 128) * 0.8$$

$$R = \text{approx. } 5,469 \text{ bits per second (bps)}$$

2. Time Measurement (100-byte Payload)

We measured the time required to transmit a fixed payload of 100 bytes to determine the efficiency of the link.

- Payload Size: 100 bytes = 800 bits
- Measured Transmission Time (T): 185ms (0.185s)

Theoretical Minimum Time (Th): This is the time it would take if there were zero overhead.

$$\text{Th} = \text{Total Bits} / R$$

$$\text{Th} = 800 / 5469 \Rightarrow \text{approx. } 146\text{ms}$$

3. Effective Bitrate Verification

The effective bitrate (Re) represents the actual speed of user data transmission in the real world, accounting for protocol overhead.

Formula:

$Re = \text{Total Bits} / \text{Time Measured}$

Calculation:

$Re = 800 / 0.185 \Rightarrow \text{approx. } 4,324\text{bps}$

4.Ten Measurements

14:46:14.436 -> --- SYSTEM START --- 14:46:14.436 -> ---

TEST: Sending 100 Bytes --- 14:46:14.583 -> --- BITRATE

VERIFICATION REPORT --- 14:46:14.583 -> 1. Payload

Size: 100 bytes (800 bits)

14:46:14.619 -> 2. Measured Time: 181.00 ms

14:46:14.619 -> 3. Theoretical Rate: 5468.75 bps

14:46:14.619 -> 4. Effective Rate: 4419.89 bps

14:46:14.619 -> 5. Efficiency: 80.82 %

14:46:14.619 -> -----

14:46:14.619 ->

14:46:16.470 -> Update Beacon 1: Dist=0.58m

14:46:18.521 -> Update Beacon 2: Dist=1.47m

14:46:24.471 -> Update Beacon 0: Dist=1.17m

14:46:24.471 -> -----

14:46:24.471 -> INPUTS: d0=1.17m | d1=0.58m | d2=1.47m

14:46:24.471 -> >>> CALCULATED POS: X=4.06, Y=3.95

14:46:24.471 -> -----

14:46:26.505 -> Update Beacon 1: Dist=0.54m

14:46:26.552 -> -----

14:46:26.552 -> INPUTS: d0=1.17m | d1=0.54m | d2=1.47m

14:46:26.552 -> >>> CALCULATED POS: X=4.07, Y=3.95

14:46:26.552 -> -----

14:46:28.588 -> Update Beacon 2: Dist=1.47m

14:46:28.588 -> -----

14:46:28.588 -> INPUTS: d0=1.17m | d1=0.54m | d2=1.47m **14:46:28.588 -> >>> CALCULATED POS: X=4.07, Y=3.95**

14:46:28.588 -> -----

14:46:34.519 -> Update Beacon 0: Dist=1.17m

14:46:34.519 -> -----

14:46:34.519 -> INPUTS: d0=1.17m | d1=0.54m | d2=1.47m

14:46:34.519 -> >>> CALCULATED POS: X=4.07, Y=3.95

14:46:34.519 -> -----

14:46:36.586 -> Update Beacon 1: Dist=0.54m

14:46:36.586 -> -----

14:46:36.586 -> INPUTS: d0=1.17m | d1=0.54m | d2=1.47m

14:46:36.586 -> >>> CALCULATED POS: X=4.07, Y=3.95

14:46:36.586 -> -----

14:46:38.637 -> Update Beacon 2: Dist=1.26m

14:46:38.637 -> -----

14:46:38.637 -> INPUTS: d0=1.17m | d1=0.54m | d2=1.26m

14:46:38.637 -> >>> CALCULATED POS: X=4.07, Y=3.99

14:46:38.637 -> -----

14:46:44.584 -> Update Beacon 0: Dist=1.36m

14:46:44.584 -> -----

14:46:44.584 -> INPUTS: d0=1.36m | d1=0.54m | d2=1.26m

14:46:44.584 -> >>> CALCULATED POS: X=4.10, Y=4.02

14:46:44.584 -> -----

14:46:46.646 -> Update Beacon 1: Dist=0.63m

14:46:46.646 -> -----

14:46:46.646 -> INPUTS: d0=1.36m | d1=0.63m | d2=1.26m

14:46:46.646 -> >>> CALCULATED POS: X=4.09, Y=4.02 14:46:46.646 -> -----

14:46:48.687 -> Update Beacon 2: Dist=1.08m

14:46:48.687 -> -----

14:46:48.687 -> INPUTS: d0=1.36m | d1=0.63m | d2=1.08m

14:46:48.687 -> >>> CALCULATED POS: X=4.09, Y=4.04

14:46:48.687 -> -----

14:46:54.605 -> Update Beacon 0: Dist=1.17m

14:46:54.653 -> -----

14:46:54.653 -> INPUTS: d0=1.17m | d1=0.63m | d2=1.08m

14:46:54.653 -> >>> CALCULATED POS: X=4.06, Y=4.01